

GOSFORD HIGH SCHOOL



Year 12

2011

HSC

MATHEMATICS EXTENSION I

Assessment Task #2

Time Allowed: 90 minutes+5 minutes reading time

Instructions:

- Start each question on a new sheet of paper.
- Attempt questions 1-4.
- Board approved calculators may be used.
- Write using black or blue pen.
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown in every question.

QUESTION 1: (16 Marks)

	Marks
a. A(1,0) and B(2,4) are two points on the number plane. Find the co-ordinates of the point P that divides the interval AB externally in the ratio 3:1	2
b. Sketch the region on the number plane that satisfies $y \leq \sqrt{4 - x^2}$	2
c. Find the number of ten letter arrangements that can be made from the letters of the word 'MOOLOOLABA'.	1
d. i. Find the exact values of the gradients of the tangents to the curve $y = e^x$ at the points where $x = 0$ and $x = 1$.	1
ii. Hence, find the acute angle between these tangents, correct to the nearest degree.	2
e. Solve:	2
$\frac{5}{ 2x - 1 } \leq 1$	
f. Evaluate $\lim_{x \rightarrow 0} \frac{3x}{\sin 4x}$	1

Question 1 Continued:**Marks**

- g. A meeting room contains a round table surrounded by twelve chairs. If a committee of twelve includes five women, how many arrangements around the table are there in which all the women sit together? 2
- h. Use one iteration of Newton's method to find an approximation, to the root of the equation: 3

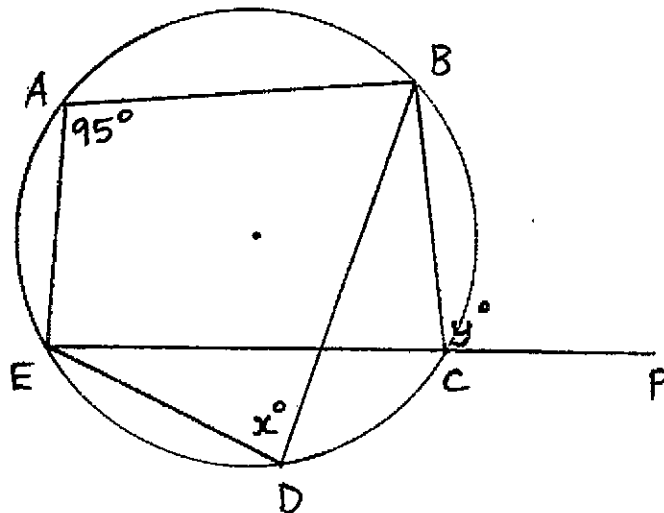
$$x \ln x - 2x = 0, \quad \text{near } x = 7$$

(Answer correct to 3 significant figures.)

End of Question 1

QUESTION 2: (18 Marks) Use a new sheet of paper

- | | Marks |
|--|-------|
| a. i. Write down the expansion of $\sin(A - B)$ | 1 |
| ii. Hence, find the exact value of $\sin \frac{\pi}{12}$
<i>(Express your answer with a rational denominator.)</i> | 2 |
| b. Find ALL values of θ for which $\sec \theta = \frac{2}{\sqrt{3}}$ | 1 |
| c. By factorising first, give the general solution, in terms of π , to the equation
$\sqrt{3} \tan^2 \theta - \sqrt{3} \tan \theta - \tan \theta + 1 = 0$ | 2 |
| d. Find the value of the pronumerals, giving reasons for your answers: | 2 |



Question 2 continued:**Marks**

- e. Let $P(x) = (x + 1)(x - 3)Q(x) + a(x + 1) + b$,
where $Q(x)$ is a polynomial and a and b are real numbers.

When $P(x)$ is divided by $(x+1)$ the remainder is 11 and
when $P(x)$ is divided by $(x - 3)$ the remainder is 1.

- i. What is the value of b ?

1

- ii. What is the remainder when $P(x)$ is divided by $(x+1)(x - 3)$

2

- f. If α, β and γ are the roots of the equation:

2

$$x^3 - x^2 + 4x - 1 = 0$$

Evaluate $(\alpha + 1)(\beta + 1)(\gamma + 1)$

- g. If the roots of the equation $x^3 - 2x^2 - x - p = 0$
are $\alpha, \alpha + 1$ and $\alpha + 3$ find the value of p

2

- h. Without the use of calculus, sketch the curve $y = \frac{x-1}{x^2-9}$

3

Clearly indicate on your sketch all asymptotes and intercepts.

End of Question 2

QUESTION 3: (18 Marks) Use a new sheet of paper

	Marks
a. Prove, by mathematical induction, that $n^3 + 2n$ is divisible by 12 for all even positive integers.	3
b. i. Show that $f(x) = x^4 - 10$ has a root between $x = 1$ and $x = 2$.	1
ii. Hence, use the method of halving the interval to show that $\sqrt[4]{10}$ lies between 1.75 and 1.875	2
c. Find $\int 2\sin^2 3x \, dx$	2
d. Use the substitution $x = u^2 - 1$ to evaluate	3
$\int_0^3 \frac{x}{\sqrt{x+1}} \, dx$	
e. Use the table of standard integrals on page 10 to show that	2
$\int_0^a \frac{1}{\sqrt{x^2+a^2}} \, dx \text{ is independent of } a$	

Question 3 continued:**Marks**

- f. The region bounded by the curve

$$y = 1 + \cos x, \text{ the } x \text{ axis, } x = 0 \text{ and } x = \pi$$

is rotated about the x axis to form a solid.

- i. Show that the volume of the solid obtained is given by:

1

$$V = \pi \int_0^{\pi} 1 + 2\cos x + \cos^2 x \, dx$$

- ii. Hence, find the exact volume of the solid.

2

- g. Use the substitution $u = x - 1$ to find

2

$$\int 3x(1 - x)^3 dx$$

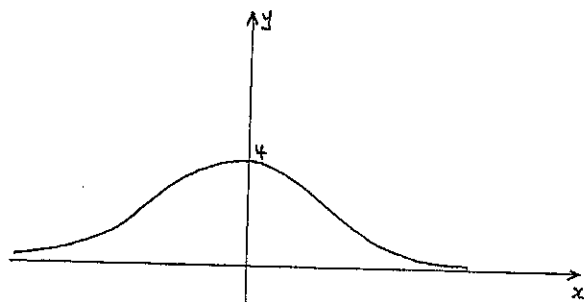
End of Question 3

QUESTION 4: (18 Marks) Use a new sheet of paper**Marks**a. Consider the function $f(x) = \log_e (x - 4)$ i. Find the inverse function $f^{-1}(x)$

1

ii. State the range of $f^{-1}(x)$

1

iii. On the same set of axes sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$, clearly indicating all asymptotes and intercepts.~~1~~ 2b. Given the sketch of $g(x) = \frac{4}{1+x^2}$ 

i. Copy this sketch.

ii. State the largest domain containing $x = -1$ for which $g(x)$ has an inverse function.

1

iii. Let $g^{-1}(x)$ be the inverse function corresponding to this restricted domain. What is the domain of $g^{-1}(x)$

1

iv. On the same set of axes in (i) sketch $y = g^{-1}(x)$

1

v. Hence, find the equation of the inverse function $g^{-1}(x)$

2

Question 4 continued:

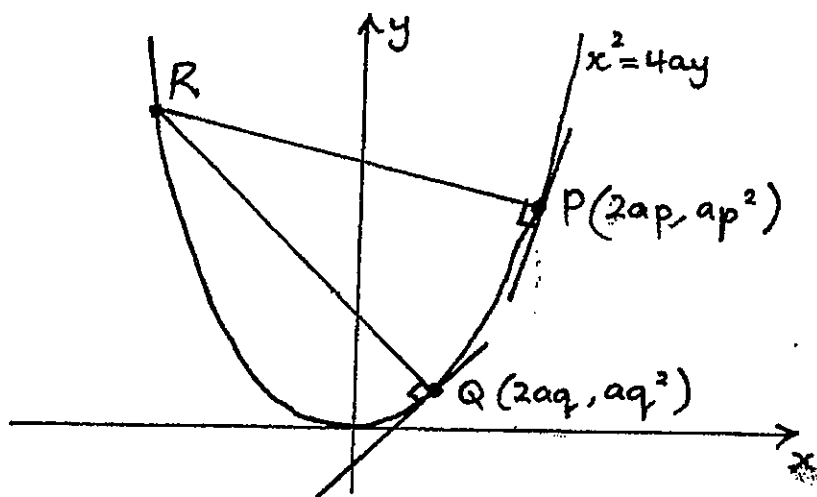
Marks

- c. The function $h(x) = \frac{x+4}{x+1}$ is defined for $x > -1$
 Let $h^{-1}(x)$ be the inverse function corresponding to the domain of $h(x)$.

Find all values of x for which $h(x) = h^{-1}(x)$

2

- d. P and Q are points on the parabola $x^2 = 4ay$ as shown in the diagram:



Normals from P and Q intersect at R. You may use the equation of the normal from P to be: $x + py = ap^3 + 2ap$ (DO NOT PROVE THIS!)

- i. Show that the normals from P and Q intersect at the point $(-apq(p+q), a(p^2 + pq + q^2 + 2))$ 3
- ii. If this point of intersection R also lies on the parabola show that $pq = 2$ 2
- iii. Hence, find the cartesian equation of the locus of the midpoint M of PQ 2

END OF TEST

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

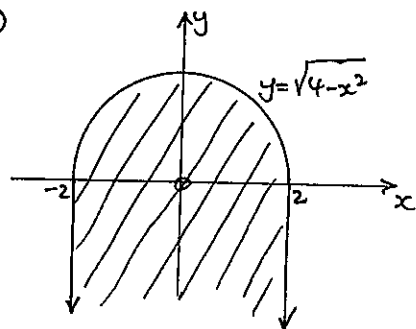
NOTE : $\ln x = \log_e x$, $x > 0$

Q1 (a) $P = \left(\frac{1x-1+3x2}{3-1}, \frac{0x-1+3x4}{3-1} \right)$

$= \left(\frac{-1+6}{2}, \frac{12}{2} \right)$

$= \left(\frac{5}{2}, 6 \right)$

(b)



(c) Arrangements = $\frac{10!}{4!2!2!}$
 $= 37800$

(d) (i) $\frac{dy}{dx} = e^x$

(when $x=0$) $\frac{dy}{dx} = 1 \rightarrow m_1$

(when $x=1$) $\frac{dy}{dx} = e \rightarrow m_2$

$\therefore$ gradients of tangents are 1 (at $x=0$) and e (at $x=1$)

(ii) using $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$
 $= \left| \frac{1 - e}{1 + e} \right|$

$\therefore \tan \theta = 0.4621 \dots$ (calc)

$\theta = 24.801 \dots$ (calc)

$\therefore$ acute angle = 25°

(e) $\frac{5}{|2x-1|} \leq 1$ CV: $x \neq \frac{1}{2}$

Consider $\frac{5}{|2x-1|} = 1$

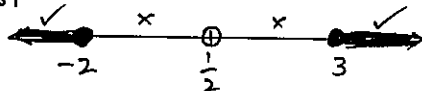
$\therefore 5 = |2x-1|$

$2x-1 = 5$ or $2x-1 = -5$

$2x = 6$ $2x = -4$

$x = 3$ $x = -2$

Test



$\therefore$ Solns: $x < -2, x > 3$

(f) $\lim_{x \rightarrow 0} \frac{3x}{\sin 4x} = \frac{3}{4}$

(g) Consider the 5 women as one group $\therefore$ 8 around a circular table = $7!$
 Ways in which women arranged = $5!$

$\therefore$ Total ways = $7! \times 5!$
 $= 604800$

(h) $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$ $\left\{ \begin{array}{l} f(x) = x \ln x - 2x \\ f'(x) = \ln x + 1 - 2 \end{array} \right.$
 $x_2 = 7 - \frac{7 \ln 7 - 14}{\ln 7 - 1}$
 $= 7.40$

Q2 (a) (i)

$\sin(A-B) = \sin A \cos B - \cos A \sin B \therefore 11 = 0 + 0 + b$

$\therefore b = 11$

(ii) Since $\sin \frac{\pi}{12} = \sin \left(\frac{\pi}{4} - \frac{\pi}{6} \right)$

$\sin \frac{\pi}{12} = \sin \frac{\pi}{4} \cos \frac{\pi}{6} - \cos \frac{\pi}{4} \sin \frac{\pi}{6}$

$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2}$

$= \frac{\sqrt{3} - 1}{2\sqrt{2}}$

$= \frac{\sqrt{3}-1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$

$= \frac{\sqrt{6} - \sqrt{2}}{4}$

$\cos \theta = \frac{\sqrt{3}}{2}$

(b) $\therefore \theta = 2\pi n \pm \frac{\pi}{6}$

(c) $\sqrt{3} \tan^2 \theta - \sqrt{3} \tan \theta - \tan \theta + 1 = 0$

$\sqrt{3} \tan \theta (\tan \theta - 1) - (\tan \theta - 1) = 0$

$(\sqrt{3} \tan \theta - 1)(\tan \theta - 1) = 0$

$\therefore \tan \theta = \frac{1}{\sqrt{3}}$ or $\tan \theta = 1$

$\therefore \theta = n\pi + \frac{\pi}{6}$ or $\theta = n\pi + \frac{\pi}{4}$

(d) $x = 85$ (opposite $\angle$'s in a cyclic quadrilateral are supplementary)

Similarly $\hat{BCE} = 85^\circ$

$\therefore y = 95$ (adjacent supplementary $\angle$'s)

(e) (i) $P(-1) = 11$

$\sin(A-B) = \sin A \cos B - \cos A \sin B \therefore 11 = 0 + 0 + b$

$\therefore b = 11$

(ii) $P(3) = 1$

$\therefore 1 = 0 + 4a + 11$

$-10 = 4a$

$-\frac{5}{2} = a$

$\therefore$ when $P(x)$ is divided by $(x+1)(x-3)$

the remainder is $a(x+1) + b$

i.e. $-\frac{5}{2}(x+1) + 11$

$= -\frac{5}{2}x - \frac{5}{2} + 11$

$= -\frac{5x}{2} + \frac{17}{2}$

(f) $\alpha + \beta + \gamma = 1$

$\alpha\beta + \alpha\gamma + \beta\gamma = 4$

$\alpha\beta\gamma = 1$

$(\alpha+1)(\beta+1)(\gamma+1) = (\alpha\beta + \alpha + \beta + 1)(\gamma+1)$

$= \alpha\beta\gamma + \alpha\beta + \alpha\gamma + \alpha$

$+ \beta\gamma + \beta + \gamma + 1$

$= \alpha\beta\gamma + (\alpha\beta + \alpha\gamma + \beta\gamma)$

$+ (\alpha + \beta + \gamma) + 1$

$= 1 + 4 + 1 + 1$

$= 7$

Q2(g)

$$x + (x+1) + (x+3) = -\frac{b}{a}$$

$$3x + 4 = 2$$

$$3x = -2$$

$$x = -\frac{2}{3}$$

$$\therefore \text{If } P(x) = x^3 - 2x^2 - x - p$$

$$P(-\frac{2}{3}) = (-\frac{2}{3})^3 - 2(-\frac{2}{3})^2 - (-\frac{2}{3}) - p$$

$$= -\frac{8}{27} - \frac{8}{9} + \frac{2}{3} - p$$

$$\text{and since } P(-\frac{2}{3}) = 0$$

$$-\frac{14}{27} - p = 0$$

$$\therefore p = -\frac{14}{27}$$

$$(h) y = \frac{x-1}{x^2-9}$$

Vertical Asymptotes: $x = \pm 3$

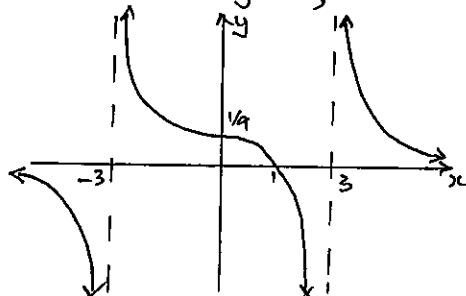
Horizontal Asymptotes:

$$\text{as } x \rightarrow \infty, y \rightarrow 0^+$$

$$x \rightarrow -\infty, y \rightarrow 0^-$$

$$\text{Intercepts: } x=0, y=\frac{1}{9}$$

$$y=0, x=1$$



Q3 Step 1: to prove true for $n=2$

$$\text{i.e. } 2^3 + 2(2) = 12 \text{ which is divisible by 12}$$

$$\therefore \text{true for } n=2$$

Step 2 Assume true for $n=2k$

(where k is a positive integer)

$$\text{i.e. } \frac{(2k)^3 + (2k)2}{12} = M$$

(where M is a positive integer)

$$\text{i.e. } 8k^3 + 4k = 12M$$

Step 3 to prove true for $n=2k+2$

$$\text{i.e. } (2k+2)^3 + 2(2k+2) \text{ is divisible by 12}$$

$$(2k+2)^3 + 2(2k+2)$$

$$= 8(k+1)^3 + 4(k+1)$$

$$= 8(k^3 + 3k^2 + 3k + 1) + 4k + 4$$

$$= 8k^3 + 24k^2 + 24k + 8 + 4k + 4$$

$$= 8k^3 + 24k^2 + 28k + 12$$

$$= 8k^3 + 4k + 24k^2 + 24k + 12$$

$$= 12M + 24k^2 + 24k + 12$$

$$= 12(M + 2k^2 + 2k + 1)$$

which is divisible by 12

$\therefore$ if true for $n=2k$ then true for $n=2k+2$

$\therefore$ true by induction for all even positive integers.

$$(b) (i) f(1) = -9 < 0$$

$$f(2) = 6 > 0$$

$\therefore$ there is a root between

$$x=1 \text{ and } x=2$$

$$(ii) \text{ if } x^4 - 10 = 0$$

$$x^4 = 10$$

$$\therefore x = \sqrt[4]{10}$$

is a root of $f(x) = 0$

Since a root lies between

$$x=1 \text{ and } x=2$$

$$\text{Let } x_1 = 1.5$$

$$f(1.5) < 0$$

$\therefore$ Root lies between $x=1.5$ and $x=2$

$$\text{Let } x_2 = 1.75$$

$$f(1.75) < 0 \therefore \text{root lies}$$

between $x=1.75$ and $x=2$

$$\text{Let } x_3 = 1.875$$

$$f(1.875) > 0$$

$\therefore$ Root lies between

$$x=1.75 \text{ and } x=1.875$$

as required.

$$(c) \int 2 \sin^2 3x \, dx$$

$$= 2 \int \frac{1}{2} (1 - \cos 6x) \, dx$$

$$= \int 1 - \cos 6x \, dx$$

$$= x - \frac{1}{6} \sin 6x + C$$

$$(d) \int_0^3 \frac{x}{\sqrt{x+1}} \, dx \quad x = u^2 - 1$$

$$\frac{dx}{du} = 2u$$

$$dx = 2u \, du$$

$$x=3 \quad u=2$$

$$x=0 \quad u=1$$

$$= \int_1^2 \frac{u^2 - 1}{\sqrt{u^2 + 1}} \cdot 2u \, du$$

$$= \int_1^2 \frac{2u^3 - 2u}{\sqrt{u^2}} \, du$$

$$= \int_1^2 (2u^2 - 2) \, du$$

$$= \left[\frac{2u^3}{3} - 2u \right]_1^2$$

$$= \left(\frac{16}{3} - 4 \right) - \left(\frac{2}{3} - 2 \right)$$

$$= \frac{10}{3} - \left(-\frac{4}{3} \right)$$

$$= 2\frac{2}{3}$$

$$(e) \int_0^a \frac{1}{\sqrt{x^2 + a^2}} \, dx$$

$$= \left[\ln |x + \sqrt{x^2 + a^2}| \right]_0^a$$

$$= \ln |a + \sqrt{2a^2}| - \ln |\sqrt{a^2}|$$

$$= \ln \left| \frac{a + \sqrt{2}a}{a} \right|$$

$$= \ln |1 + \sqrt{2}| = \ln(1 + \sqrt{2})$$

which is independent of a

$$(f) (i) V = \pi \int_0^{\pi} y^2 dx$$

$$= \pi \int_0^{\pi} (1 + \cos x)^2 dx$$

$$= \pi \int_0^{\pi} 1 + 2\cos x + \cos^2 x dx$$

as required

$$(ii) V = \pi \int_0^{\pi} 1 + 2\cos x + \frac{1}{2}(1 + \cos 2x) dx$$

$$= \pi \left[x + 2\sin x + \frac{1}{2}x + \frac{1}{4}\sin 2x \right]_0^{\pi}$$

$$= \pi \left[\left(\pi + 0 + \frac{\pi}{2} + 0 \right) - 0 \right]$$

$$= \frac{3\pi^2}{2} \text{ units}^3$$

$$(g) \int 3x(1-x)^3 dx$$

$$u = x-1 \Rightarrow x = u+1$$

$$\frac{du}{dx} = 1$$

$$du = dx$$

$$= -3 \int u^4 + u^3 du$$

$$= -3 \left(\frac{u^5}{5} + \frac{u^4}{4} \right) + C$$

$$= -3 \left(\frac{(x-1)^5}{5} + \frac{(x-1)^4}{4} \right) + C$$

$$Q4 (a) (i) y = \log_e(x-4)$$

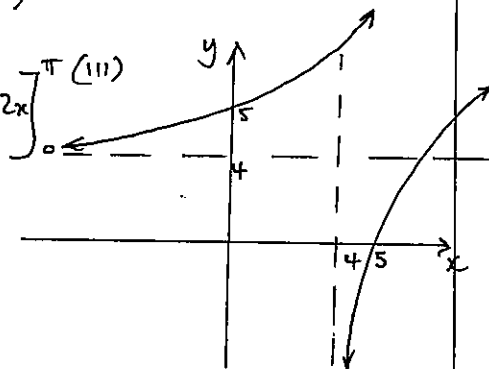
$$x = \log_e(y+4)$$

$$e^x = y+4$$

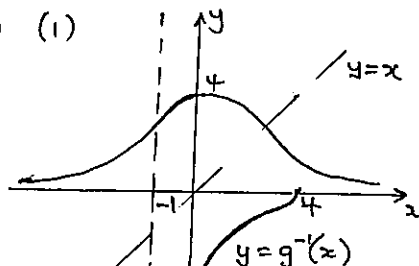
$$e^x + 4 = y$$

(ii) Range of $f^{-1}(x)$

$$: y > 4$$



(b) (i)



(ii)

Apply horizontal line test
largest restricted domain
is $x \leq 0$

(iii) Domain of $g^{-1}(x)$
corresponds to range of $g(x)$

$$\text{i.e. } 0 < x \leq 4$$

(iv) on above sketch

$$(v) y = \frac{4}{1+x^2}$$

$$x = \frac{4}{1+y^2}$$

$$1+y^2 = \frac{4}{x}$$

$$y^2 = \frac{4}{x} - 1$$

$$y^2 = \frac{4-x}{x}$$

$$y = \pm \sqrt{\frac{4-x}{x}}$$

Since y is negative.

$$y = -\sqrt{\frac{4-x}{x}}$$

(c) Point of intersection
 $h(x)$ and $h^{-1}(x)$

Solve $y = h(x)$ and $y = x$

$$\therefore \frac{x+4}{x+1} = x$$

$$x+4 = x^2+x$$

$$0 = x^2 - 4$$

$$\pm 2 = x$$

(Since $x > -1$) $x = 2$
is the only solution.

(d) Normal at P:

$$x + py = ap^3 + 2ap \quad - (1)$$

Normal at Q:

$$x + qy = aq^3 + 2aq \quad - (2)$$

$$\textcircled{1} - \textcircled{2} : (p-q)y = a(p^3 - q^3) + 2a(p-q)$$

$$y = a(p^2 + pq + q^2) + 2a$$

$$= a(p^2 + pq + q^2 + 2)$$

Sub in $\textcircled{1}$:

$$x + ap(p^2 + pq + q^2 + 2) = ap^3 + 2ap$$

$$x = ap^3 + 2ap - ap^3 - apq - apq^2 - 2ap$$

$$= -apq(p+q)$$

$\therefore$ Pt of intersection: $(-apq(p+q), a(p^2 + pq + q^2 + 2))$
as required.

(ii) If R lies on $x^2 = 4ay$

$$\therefore [-apq(p+q)]^2 = 4a[a(p^2 + pq + q^2 + 2)]$$

$$a^2 p^2 q^2 (p+q)^2 = 4a^2 (p^2 + pq + q^2 + 2)$$

$$p^2 q^2 (p+q)^2 = 4(p^2 + pq + q^2 + 2)$$

$$p^2 q^2 (p+q)^2 = 4(p^2 + 2pq + q^2) + 8 - 4pq$$

$$p^2 q^2 (p+q)^2 = 4(p+q)^2 + 4(2-pq)$$

$$p^2 q^2 (p+q)^2 - 4(p+q)^2 - 4(2-pq) = 0$$

$$(p+q)^2 [p^2 q^2 - 4] - 4(2-pq) = 0$$

$$(p+q)^2 (pq-2)(pq+2) - 4(2-pq) = 0$$

$$(pq-2) [(p+q)^2 (pq+2) + 4] = 0$$

$$\therefore pq = 2 \quad (\text{or show LHS = RHS})$$

$$(iii) M = (a(p+q), \frac{ap^2 + aq^2}{2})$$

$$\therefore x = a(p+q) \quad - (1)$$

$$y = \frac{a}{2}(p^2 + q^2) \quad - (2)$$

$$\text{From } \textcircled{1} \quad \frac{x}{a} = p+q \quad \textcircled{2} \Rightarrow \frac{y}{a} = \frac{a}{2}(p+q)^2 - 2pq$$

$$\textcircled{2} \Rightarrow y = \frac{a}{2} \left(\left(\frac{x}{a} \right)^2 - 4 \right) \therefore \frac{2y}{a} = \frac{x^2}{a^2} - 4$$